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Lab 3 - Architecture Selection and Component Modelling

Architecture Selection

I chose layered architecture for the Centralized Audit Trail Compliance Engine because the system has clear stages for receiving audit logs, processing and masking sensitive data, querying records and storing audit data. The layered structure makes these responsibilities easier to separate and understand

Reason 1 - Clear separation of responsibilities

The Audit Log Ingestion and PII Masking Service handle the processing side of the system while the Audit Query Service and Immutable Audit Export Service handle user operations. Audit Storage is kept in the data layer. This makes the design easier to maintain and reduces confusion between processing and storage responsibilities

Reason 2 - Suitable for a centralized compliance system

The assigned system is designed as a centralized audit trail platform used by a Compliance Officer and a Security Auditor. A layered structure fits this type of system because the important audit functions can work through defined interfaces while the storage part remains separate from the service components

Security Advantage

A major security benefit is that PII masking is placed before the storage stage. Sensitive values such as credit card numbers can be identified and masked before the audit record is saved. The export service also supports SHA-256 integrity hashing so the generated audit export can be checked for tampering

Performance Benefit

The separate Audit Query Service allows query and filtering operations to be handled independently from log ingestion and PII masking. This is useful for the requirement to support complex filtering across up to 10 million historical records in under one second

Conclusion

Overall, layered architecture is a practical choice for this system because it keeps the main responsibilities separate while supporting the security and performance requirements of the audit platform